Final Project

1 Introduction

The main objective of this course is to prepare you to apply data analytics techniques and AI algorithms to real-world tasks. The final project aims to help you get started in this direction. You will be performing the following tasks:

- Try out existing techniques and algorithms to a real-world task.
- Write a short report analyzing the result.

The final project is a good chance for you to experiment, think, play, and hopefully have fun.

2 Background

Financial market predictions utilize historical data to anticipate future stock prices and market trends. Traditionally, these predictions have focused on the statistical analysis of quantitative factors, such as stock prices, trading volumes, inflation rates, and changes in industrial production. Recent advancements in machine learning motivate the integrated financial analysis of both social media data and numerical factors.

3 Data

This dataset uniquely combines time series news and stock prices, providing a valuable resource for financial market analysis.

- The “Tweet” dataset contains over 3 million unique tweets with their information such as tweet id, author of the tweet, post date, the text body of the tweet, and the number of comments, likes, and retweets of tweets matched with the related company.
- The “Stock_price” dataset contains numerical stock price data collected from Yahoo Finance’s API.

4 Task

4.1 Main task

Stock price movement prediction

- Regarding it as a binary classification task (Class 1: the stock price moves up; Class 2: the stock price remains unchanged or moves down)
- Using information collected in period t (e.g., stock price and tweets) to predict the stock price movement in period (t+1) and period (t+7)
- Using machine learning, deep learning or LLM approaches
- You are required to use textual data in the “Tweet” dataset.
- Performance measurement: precision, recall, F1 score
- You need to submit code and any additional data/features you used for analysis.
- You need to submit a final PDF report, including project background, objectives, model design, results analysis, etc. No more than 10 pages.

Hint:

- Textual and numerical data integration in the dataset may enhance model functionality and performance.
- You can consider social media engagement data that reflects the influence of tweets, including the number of comments, the number of likes, the number of retweets, and sentiment of tweets.
- You can consider financial control variables, which are known to have predictive power for stock returns.

4.2 Optional: Market simulation

To show the economic value of your model, you can simulate real market trading. The basic idea is to mimic the behavior of a trader who make decision based on your model. The trader has $10,000 as seed money. If the model forecasts that an individual stock price will move up the next day, the trader will invest in $10,000 worth of that stock at the opening price and sells the stock at the closing price. If the model forecasts that an individual stock price will move down, the trader will take no action. You can calculate the average profits over the individual stocks that are presented in the dataset, e.g., average profit on day t+1 and average profit on day t+7.

Appendix

Checklist of Machine Learning Projects

- Frame the problem and look at the big picture.
    - Define the objective in business terms.
    - How will your solution be used?
    - What are the current solutions/workarounds (if any)?
    - How should you frame this problem (supervised/unsupervised, online/offline, etc.)
    - Is human expertise available?
- Get the data.
    - Convert the data to a format you can easily manipulate (without changing the data itself).
    - Ensure sensitive information is deleted or protected (e.g., anonymized).
    - Check the size and type of data (time series, sample, geographical, etc.).
    - Sample a test set, put it aside, and never look at it (no data snooping!).
- Explore the data to gain insights.
    - Visualize the data.
    - Study the correlations between attributes.
- Prepare the data to better expose the underlying data patterns to Machine Learning algorithms.
    - Write functions for all data transformations you apply
    - Data cleaning
    - Feature selection (optional)
    - Feature engineering
    - Feature scaling
- Explore many different models and short-list the best ones.

    - Train many quick and dirty models from different categories (e.g., linear, Bayes, Random Forests, neural net, etc.) using standard parameters.
    - Measure and compare their performance.
    - Analyze the most significant variables for each algorithm.
    - Have a quick round of feature selection and engineering.
    - Have one or two more quick iterations of the five previous steps.
    - Short-list the top two to four most promising models, preferring models that make different types of errors.
- Fine-tune your models and combine them into a great solution.
    - Fine-tune the hyperparameters using cross-validation.
    - Try Ensemble methods.
    - Once you are confident about your final model, measure its performance on the test set to estimate the generalization error.
- Present your solution.
    - Document what you have done.
    - Create a nice presentation.
    - Explain why your solution achieves the business objective.
    - Don't forget to present interesting points you noticed along the way.
    - Ensure your key findings are communicated through beautiful visualizations or easy-to-remember statements (e.g., "the median income is the number-one predictor of housing prices").